Dennis G. Pezan

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EDUCATION

United States Military Academy at West Point — West Point, NY
B.S., Operations Research (Finance Specialization), Minor in Artificial Intelligence — Class of 2029

- CQPA: 3.571 | Dean's List, Fall 2025 and Spring 2026 | Secret Clearance
- Research Project: Threat-Adaptive Interdiction of Taiwan's Crude Oil Supply (Advisor: LTC Matthew Webb)

RESEARCH EXPERIENCE

MIT Center for Transportation & Logistics – USMA Research — West Point, NY
Undergraduate Researcher, Network Interdiction for Contested Logistics (Advisor: LTC Matthew Webb) — 2026–Present

- Project: Threat-Adaptive Interdiction of Taiwan's Crude Oil Supply; extends LTC Webb's MIT Center for Transportation & Logistics doctoral research on reinforcement learning for maximum-flow network interdiction
- Investigate how to optimally attack and defend military logistics networks under adversarial interdiction, applying network optimization, game theory, and Markov decision processes
- Built a 57-node, 79-edge open-source model of Taiwan's crude oil import network (four sources, major straits and canals, three ports) with sourced capacities
- Designed a fixed-budget adversary action set of seven action types, each with a cost and success probability anchored on historical events
- Connected the network to a reinforcement-learning interdiction environment and verified it against an independent max-flow calculation; identified and fixed a targeting-filter issue that could hide the Strait of Hormuz

Transformation Decision Analysis Center (TDAC), U.S. Army — Fort Lee, VA
Research Intern, Modeling & Analysis Division (AIAD; Mentor: MAJ Erica Mikelson) — Summer 2026

- Redesigned the reward function for a proximal policy optimization (PPO) reinforcement-learning agent that allocates resupply in Logistics Battle Command, the Army's stochastic sustainment simulation; trained and evaluated the agent in Python (PyTorch)
- Built two diagnostic tools that exposed four reward flaws, including an agent that outscored every baseline while starving the priority unit during engagements
- Retrained agent more than tripled the share of requests filled and outperformed a hand-coded priority-aware baseline across 101 random seeds
- Presented findings to Branch Chief LTC Ryan Miller at Fort Lee and, by call, to TDAC's AI and reinforcement-learning researchers (LTC Joseph Pedersen) at the Naval Postgraduate School in Monterey, CA

Department of Mathematical Sciences, USMA — West Point, NY
Independent Study in Machine Learning (MA289), Advisor: LTC Patrick Kuiper — 2025–2026

- Designed and completed an independent study of decision trees, bagging, and random forests, and presented the findings to officers
- Built an interactive random forest research site with a live classifier demonstration and hyperparameter explorer

PROJECTS

Tree-Based Methods Research Site – dennis-pezan.com/ma289.html — 2026

- Trained and evaluated decision trees, bagging, and random forests on the Titanic dataset in Python (scikit-learn) on an identical 80/20 split
- Computed pairwise tree correlation, overfitting gaps, and feature importance to show how ensemble decorrelation reduces variance; deployed via GitHub and Vercel

CFT Packing List – dennis-pezan.com/cft.html — 2026

- JavaScript web tool that helps cadets organize and track gear for Cadet Field Training; saves progress in the browser and deploys on Vercel from GitHub

Reward Engineering Case Study – dennis-pezan.com/rl-reward.html — 2026

- Public write-up diagnosing and correcting reward hacking in a simulated shipping network; requests met rose from 24% to 81%

Mounger Writing Center Website Redesign – dennis-pezan.com/mwc-redesign.html — 2026

- Independent redesign concept (in progress) for West Point's writing center site, built in Next.js, TypeScript, and Tailwind CSS with a custom WebGL ink shader and editable content files for staff

Professional Development

Sea to Shining Sea Domestic AIAD — California and Colorado
Defense Technology and Innovation — Summer 2026

- Held in-depth discussions with leaders at Palantir, Stanford's Gordian Knot Center (Col. (Ret.) Joseph Felter, Jack Faherty), OpenAI, Apple, and Meta Reality Labs (CTO Andrew Bosworth) on how defense technology is developed, procured, and fielded
- Topics: startup barriers to Army adoption, agentic AI and large language models for defense, supply-chain risk, and AI regulation
- Observed how the 4th Infantry Division employs and gives feedback on these systems under Next Generation Command and Control; visited U.S. Northern Command, NORAD, and the Cheyenne Mountain Complex

Honors & Awards

Dean's List, United States Military Academy — Fall 2025, Spring 2026
Stokes Writing Fellowship, United States Military Academy — 2026–Present

- Endowed fellowship funding research travel, expert mentorship, and publication of original research

Norwegian Foot March Badge, Norwegian Armed Forces — 2026
Sea to Shining Sea Fellowship, United States Military Academy — 2026–Present

- Cross-disciplinary fellowship engaging cadets with senior leaders in national security, government, finance, and technology

Leadership & Service

Center for Enhanced Performance, USMA — West Point, NY
Cadet-in-Charge, MA103 Data-Driven Modeling Tutoring — 2026–Present

- Lead the full tutor staff for MA103, a core mathematics course with 1,010 cadets enrolled; direct tutoring coverage for every major graded event
- Built a study site (dennis-pezan.com/ma103-tutoring.html) with two full-length practice exams, a mirror of a graded gate, and a 59-drill reference card, each with an answer key
- Certified tutor in mathematics, chemistry, and psychology since 2025

Mounger Writing Center, USMA – *Writing Consultant* — West Point, NY | 2026–Present

- Selected as a Stokes Writing Fellow to coach cadets on strengthening the rhetoric and argument of their writing

CLD-STEM Outreach Program, USMA – *Member* — West Point, NY | 2025–Present

- Bring hands-on STEM projects to students, reaching 800+ students across the Chicagoland and Greater Los Angeles areas

Cadet Ambassador Program, USMA – *Cadet Ambassador* — West Point, NY | 2025–Present

- Represent West Point in outreach to prospective cadets and help foster a welcoming environment for future applicants

Finance Club, USMA – *Member* — West Point, NY | 2025–Present

- Engage with industry executives from companies including Palantir, Uber, and Meta

Recycle & Reboot – *Volunteer Senior Publicist* — Plainfield, IL | 2024

- Led outreach to secure donated technology; helped refurbish and distribute 100+ computers to underserved communities

Illinois Premier Boys State – *Candidate* — Illinois | 2024

- Elected Alderman and Township Supervisor; advanced to the Lieutenant Governor primary through competitive public speaking

Professional Experience

McDonald's — Plainfield, IL
Department Manager — September 2022 – July 2024

- Promoted at age 16 as the region's youngest department manager; sustained 90%+ customer satisfaction scores
- Scheduled 80+ employees weekly with 95% on-time accuracy; oversaw $20,000+ in weekly supply procurement
- Owned the store's labor metric, holding it at or below 23.5% month over month

GrayStone AI Solutions — Remote
Intern — June – July 2024

- Built Python and Excel tools to support financial planning and analysis workflows

Skills

Technical: Python (PyTorch, scikit-learn), SQL, Java, JavaScript/TypeScript (Next.js) | **Methods:** reinforcement learning, network optimization, machine learning
Certifications: JPMorgan Chase Software Engineering Job Simulation (2026); Microsoft Fabric Data Architecture (2026)
Languages: Spanish (fluent) | **Clearance:** Secret